
MATH6400: Quiz #6**Due:** March 27, 2026 @ 23:59**Fermat's Legendary Marginalia**

In about 1625, Fermat studied and was inspired by the work in a then-new translation of Diophantus' *Arithmetica*. In 1670, shortly after Fermat died, his son published a version of Diophantus' book that included the marginalia of Fermat, which included the following:

If an integer n is greater than 2, then $a^n + b^n = c^n$ has no solutions in non-zero integers a , b , and c . I have a truly marvelous proof of this proposition which this margin is too narrow to contain.

While he was alive, Fermat challenged several mathematicians to prove the non-existence of a Pythagorean triangle with square area; a result which he knew to be true, but did not publish a proof himself. However, he wrote a proof in his copy of Diophantus' *Arithmetica*. This is the only known complete proof given by Fermat of one of the many results attributed to him.

Fermat's proof of the following theorem used a method now-called *infinite descent* of which Fermat said "this method will enable extraordinary developments in the theory of numbers".

Theorem: (*Fermat's Rational Right Triangle Theorem*) *There does not exist a right triangle with sides all of rational length and with area equal to the square of a positive integer.*

Infinite descent uses *reductio ad absurdum** and an induction-like argument that produces an apparently non-terminating decreasing sequence of positive integers. Since there is a smallest positive integer, this contradiction forces the denial of the assumption.

Prompts:

Suppose right triangle T has sides of length a and b and hypotenuse of length c ; to T we associate the triple (a, b, c) or (b, c, a) . The triple (a, b, c) is a *primitive Pythagorean triple*, or T is a *primitive Pythagorean triangle* if a , b , and c are positive integers which have no common divisor.

1. Prove **Lemma 1**. *If (a, b, c) is a primitive Pythagorean triple, then one of a or b is odd and the other is even.*

Note: Therefore, the area of a primitive Pythagorean triangle is an even integer.

Solution: This lemma is forced from the definition of a primitive Pythagorean triple. Because a, b, c must be coprime, then either a or b must be odd and the other even.

Assume a, b are both even, then they share the common divisor 2. Boom.

Assume a, b are both odd and share no common divisor. Observe since we have a right triangle the Pythagorean theorem applies. Then we can see $c^2 = a^2 + b^2$. The only instance $\sqrt{a^2 + b^2} \in \mathbb{Z}$ is when $a^2 + b^2 \equiv 0 \text{ or } 1 \pmod{4}$. We can see since our values are odd

$$a = 2k + 1 \Rightarrow a^2 = (2k + 1)^2 = 4k^2 + 4k + 1 \equiv 1 \pmod{4}$$

If we add two values of this form then we see

$$a^2 + b^2 \equiv 1 + 1 \equiv 2 \pmod{4}$$

Therefore, either a or b must be even, and the other value must be odd.

2. Prove **Lemma 2**. *If there is a right triangle with sides and hypotenuse having rational number lengths and whose area is a square integer, then there is a primitive Pythagorean triangle with sides of integer length.*

*Proof by contradiction; reduction to the absurd.

Solution: Let $a, b, c \in \mathbb{Q}^+$ be the legs and hypotenuse of a right triangle, meaning $a^2 + b^2 = c^2$. We can see

$$a = \frac{p_1}{q_1}, b = \frac{p_2}{q_2}, c = \frac{p_3}{q_3}$$

Let d be the lowest common divisor of q_1, q_2, q_3 , then we can obtain the congruent triangle $\triangle ABC$ with sides and hypotenuse

$$A = da, B = db, C = dc$$

We see (A, B, C) is also a Pythagorean triple as

$$A^2 + B^2 = d^2 a^2 + d^2 b^2 = d^2 (a^2 + b^2) = d^2 c^2 = C^2$$

To find a primitive triple, let $g = \gcd(A, B, C)$ and define

$$A' = A/g, B' = B/g, C' = C/g$$

then these are all coprime and

$$A'^2 + B'^2 = \frac{A^2 + B^2}{g^2} = \frac{C^2}{g^2} = C'^2$$

therefore, (A', B', C') is a primitive Pythagorean triple.

But wait! There's more! The area of $\triangle A'B'C'$ is also a square integer. Either A' or B' is even by lemma 1, so area $= \frac{1}{2}A'B'$ is an integer. Observe

$$\frac{1}{2}A'B' = \frac{AB}{2g^2} = \frac{d^2 ab}{2g^2} = \frac{d^2 \frac{1}{2} ab}{g^2} = \frac{d^2 n^2}{g^2}$$

We can see since $\frac{d^2 n^2}{g^2}$ is an integer that $\frac{dn}{g}$ is a rational whose square is an integer. The only rational whose squares are integers are themselves integers, meaning $\frac{dn}{g} \in \mathbb{Z}$ and therefore

$$\frac{1}{2}A'B' = \left(\frac{dn}{g} \right)^2$$

is a perfect square. ■

3. Prove **Lemma 3**. If (a, b, c) is a primitive Pythagorean triple, then for some u and v , with $\gcd(u, v) = 1$, one of u, v even and the other odd, $a = 2uv$, $b = u^2 - v^2$, and $c = u^2 + v^2$. Furthermore, if $\frac{1}{2}ab$ is a square integer, then $u, v, u + v, u - v$ are square integers.

Solution: Let (a, b, c) be a primitive Pythagorean triple. Without loss of generality, assume a is even. Observe

$$a^2 = c^2 - b^2 = (c - b)(c + b)$$

Since c and b are both odd (by lemma 1), both $(c - b)$ and $(c + b)$ are even. Let

$$c - b = 2s$$

$$c + b = 2t$$

and therefore

$$a^2 = 4st \Rightarrow \left(\frac{a}{2} \right)^2 = st$$

Since $st = \left(\frac{a}{2} \right)^2$, st is a perfect square. Observe

$$\begin{aligned} c &= s + t \\ &= \frac{c - b}{2} + \frac{c + b}{2} = c \end{aligned}$$

Therefore, either s or t must be odd and the other must be even since c is odd. Any common divisor of s and t will also divide $t - s = b$ and $s + t = c$, but $b \perp c$, so $s \perp t$ also. Since $s \perp t$ and st is a perfect square, then s and t are also perfect squares.

This is a claim which necessitates a proof.

proof of a lemma in a lemma: Let p be a prime which divides s . Since $st = \left(\frac{a}{2}\right)^2 = k^2$, then p divides k^2 , so $p^2 | k^2$. Next, we show $p^2 | s$.

Since $s \perp t$, $p \perp t$. All factors of p in st come from s . Since st is a perfect square, every prime factor is raised to an even power. Since p 's contribution comes only from s , p must appear to an even power in s . Since this holds for all primes dividing s , every prime in s 's factorization appears to an even power, so s is a perfect square. The same argument can be applied to t to see both are themselves perfect squares.

So now we know s and t are perfect squares. Let $s = v^2$ and $t = u^2$. When we plug this substitution into our previously found equations, we obtained the claimed values of a, b, c :

$$a = \sqrt{4st} = \sqrt{4u^2v^2} = 2uv$$

$$b = t - s = u^2 - v^2$$

$$c = s + t = u^2 + v^2$$

Now, assume $\frac{1}{2}ab$ is a square integer. We can see

$$\frac{1}{2}ab = 2uv(u^2 - v^2)$$

$$\begin{aligned} ab &= uv(u^2 - v^2) \\ &= uv(u - v)(u + v) \end{aligned}$$

I claim $u, v, (u - v), (u + v)$ are all relatively prime. u and v are trivially coprime. u is coprime with $u - v$ and $u + v$ since any common divisor would also have to divide v , and v is also coprime with $u - v$ and $u + v$ since any common divisor would also have to divide u . Any common divisor of $u - v$ or $u + v$ must also divide $2u$ and $2v$ respectively. Since u and v are of opposite parity, $u + v$ and $u - v$ are odd and therefore are relatively prime. At this point, we can employ our *lemma in a lemma* to see that a perfect square construct as a product of values must be constructed by perfect squares, and therefore $u, v, u + v, u - v$ are all perfect squares. ■

4. Prove **Corollary 4**. With u and v from Lemma 3, (v, w, u) , with $w^2 = (u - v)(u + v)$, is a primitive Pythagorean triple.

Solution: We need to first show that this is a Pythagorean triple, and then can show it is a primitive Pythagorean triple. Lets verify that $v^2 + w^2 = u^2$:

$$v^2 + u^2 - v^2 = u^2 \quad \checkmark$$

In the last lemma, we already showed $u \perp v$. We can w is also coprime since that means some p would have to divide w , which by the construction of w means p divides both u and v^2 or v and u^2 . Therefore, all three are coprime, and the triple is of type primitive Pythagorean. ■

5. Find a primitive Pythagorean triangle with area a square integer less than $\frac{1}{2}ab = uv(u^2 - v^2)$. Conclude that there cannot be a rational right triangle with area a square integer.

Solution: We just found it in the Corollary! Observe given the primitive Pythagorean triples constructed with the same u, v ($2uv, u^2 - v^2, u^2 + v^2$) and $(v, \sqrt{u^2 - v^2}, u)$ that the following inequality exists between their areas:

$$\frac{1}{2}uv(u^2 - v^2) > \frac{1}{2}v\sqrt{u^2 - v^2}$$

6. Prove the following corollaries of Fermat's Rational Right Triangle Theorem.

(a) *There do not exist positive integers a, b , and c for which $a^4 - b^4 = c^2$.*

Solution: Assume there exists positive integers which satisfy the equation $a^4 - b^4 = c^2$ and assume we choose the values such that a is the smallest possible integer. Additionally, assume these values are coprime. We can then see this can be expressed as a Pythagorean triple:

$$c^2 + (b^2)^2 = (a^2)^2$$

Because the values are coprime, then we can use the substitutions we derived in lemma 3

$$\begin{aligned} b^2 &= 2mn \\ a^2 &= m^2 + n^2 \\ c &= m^2 - n^2 \end{aligned}$$

As shown in lemma 3, we can see $m, n, m^2 + n^2$ must be perfect related to perfect squares. Then we can perform another layer of substitutions to see

$$m = 2pq \quad n = p^2 - q^2, \quad a = p^2 + q^2$$

Using this, we can see

$$\begin{aligned} b^2 &= 2mn = 4pq(p^2 - q^2) = 4pq(p - q)(p + q) \\ \Rightarrow \left(\frac{b}{2}\right)^2 &= pq(p - q)(p + q) \end{aligned}$$

We know from lemma 3 these are all coprime and therefore each is itself a perfect square. Let $p = r^2$ and $q = s^2$ to obtain

$$r^4 - s^4 = p^2 - q^2 = (p - q)(p + q) = k^2, \text{ for some } k \in \mathbb{Z}$$

Then we see this is a new solution to $a^4 - b^4 = c^2$, since $r^2 = p \leq p^2 + q^2 = a$, we have $r < a$, contradicting the minimality of a . ■

(b) *There are no positive integers x, y, z for which $x^4 + y^4 = z^4$. (One case of "Fermat's Last Theorem").*

Solution: We can apply the previous part directly. Observe

$$x^4 + y^4 = z^4 \Rightarrow x^4 + y^4 = (z^2)^2$$

We just showed this case is impossible, and therefore there are no positive x, y, z which satisfy this equation.
